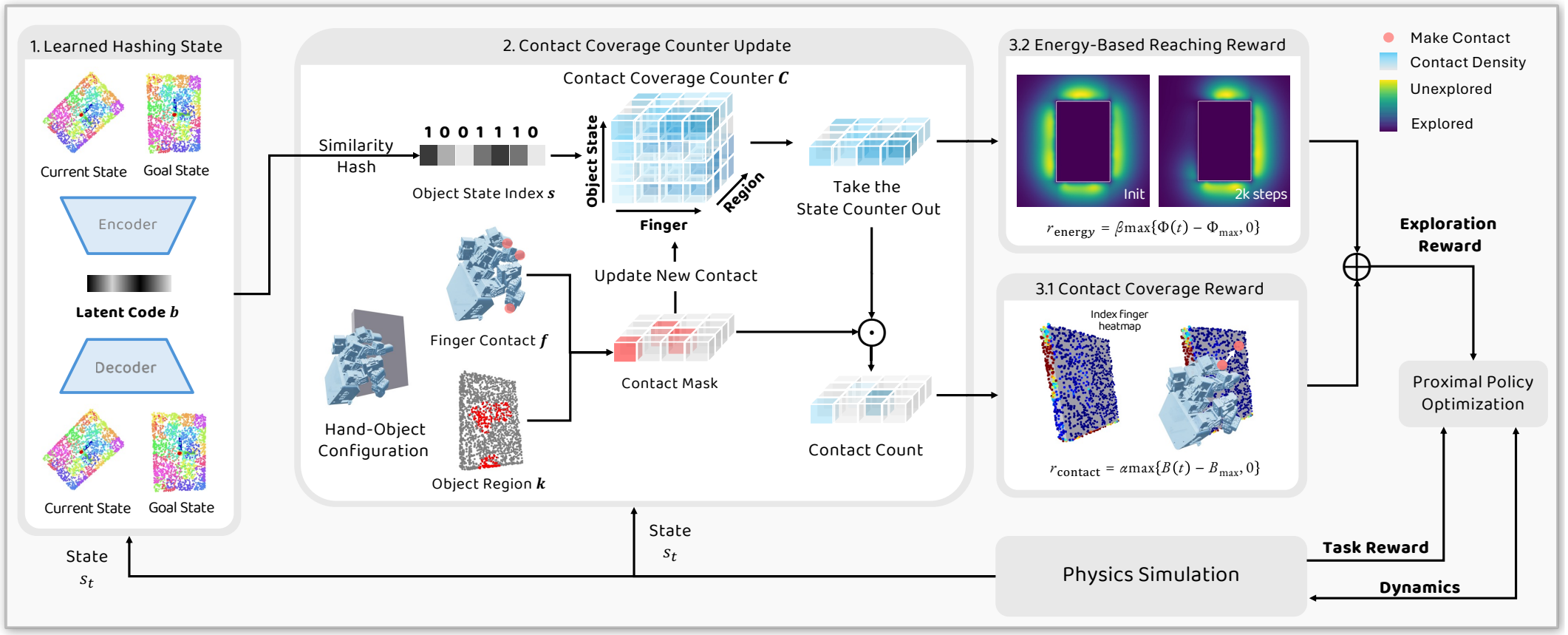
1. Learned Hashing State
Current State
Goal State
Encoder
Latent Code b
Decoder
Current State
Goal State
2. Contact Coverage Counter Update
Contact Coverage Counter C
Similarity Hash
1 0 0 1 1 1 0
Object State Index s
Object State
Region
Finger
Take the State Counter Out
Update New Contact
Finger Contact f
Hand-Object Configuration
Object Region k
Contact Mask
Contact Count
3.2 Energy-Based Reaching Reward
Init
2k steps
$r_{\text{energy}} = \beta \max\{\Phi(t) - \Phi_{\max}, 0\}$
3.1 Contact Coverage Reward
Index finger heatmap
$r_{\text{contact}} = \alpha \max\{B(t) - B_{\max}, 0\}$
Make Contact
Contact Density
Unexplored
Explored
Exploration Reward
Proximal Policy Optimization
Task Reward
Dynamics
Physics Simulation
State s_t
State s_t